A Toy Study of Layout-Preserving PDF Translation

Ada Example, Bo Sample and Cy Demo

Abstract· Translating an academic paper is easy; keeping its layout is not. Columns, floating figures, tables and numbered equations all move when the text changes length. This toy paper exists so that a layout-preserving translation pipeline can be exercised end to end without shipping anyone else's copyrighted work. It contains two columns, one full-width caption, one table and one numbered equation.

1. Introduction

Object navigation and document translation look unrelated, yet both are exercises in keeping a structured representation consistent while its contents change. In this paper we describe a deliberately small example that stresses the parts of a PDF that a naive text-replacement pipeline usually breaks: column order, figure anchoring, and table cell counts.

Our pipeline treats layout as code and language as data. The renderer never asks a language model where a paragraph should go, and the translator never sees a coordinate. Keeping that separation makes failures diagnosable: a wrong crop is a geometry bug, a wrong sentence is a prompting bug, and the two never hide each other.

The second reason is regression testing. Once the geometry is deterministic, a change to the extractor either reproduces the same boxes or it does not, and that difference can be reviewed in seconds instead of by reading a diff of coordinates.

$$C = (1/N) * \mathrm{SUM}\ p_i * d_i \qquad (1)$$

2. Method

We extract text blocks together with their bounding boxes, classify each block as title, heading, body, caption or table, and then rebuild the page in HTML. Every block is measured in a real browser before pagination, because estimating line breaks from character counts is wrong often enough to matter.

Figures are anchored to their captions: the graphic group that sits closest above a caption is almost always the figure that caption describes. Tables are anchored the other way, since table captions sit above the table in most venues.

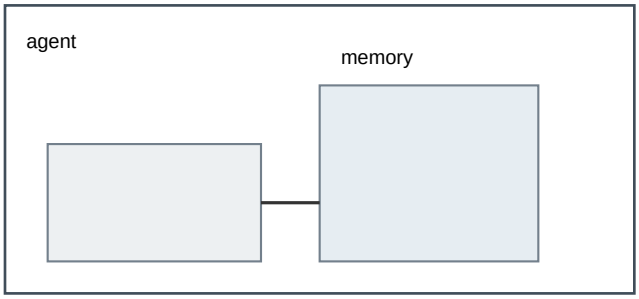

Fig. 1: A schematic figure that must stay above its own caption after reflowing.

3. Results

On the toy sample the pipeline reproduces both columns, keeps the figure above its caption, preserves the table's three columns, and leaves the equation as an image so that its indices cannot be mistyped.

Table 1: Toy results.

Method	SPL	Success
Baseline	31.2	48.5
Ours	37.4	55.8

The expected cost of a visit order is

$$E = \sum_i p_i \cdot d(o_{i-1}, o_i) \qquad (2)$$

Equation (2) is kept as a picture so that subscripts cannot be mistyped during translation.

References

[1] A. Example. Layout as code, language as data. In Proc. Toy Conf., 2026.

[2] B. Sample. Measuring blocks before pagination. arXiv:0000.00000, 2026.